

Water Tanker Booking & Delivery Management System

DBMS Mini Project — Review 1

1. Problem Identification

Scheduling Conflicts

Tankers assigned to overlapping delivery slots, resulting in missed and delayed deliveries.

No Quantity Tracking

Source, filled quantity, and delivered quantity go unrecorded – fueling billing disputes.

Ad-hoc Billing

No consistent tariff structure; invoices are informal and error-prone.

Zero Visibility

Pending bookings, delays, and complaints are invisible to management.

2. System Scope

IN SCOPE

- Customer registration & address management
- Booking creation & tanker/driver scheduling
- Filling records & delivery confirmation
- Tariff-based billing & payment tracking
- Complaint tracking & operational reporting

OUT OF SCOPE

- Route optimization and GPS navigation
- Inventory or fuel procurement management
- Payroll and HR administration
- Advanced analytics or predictive forecasting
- Third-party accounting system integration

5. Functional Requirements

1

Customer & Address Management

Register customers with unique phone/email, add multiple addresses, and retrieve full history.

2

Booking Management

Create bookings with customer, address, quantity, date, and time slot; prevent past-date bookings and track status.

3

Tanker & Driver Scheduling

List available tankers and drivers, assign one of each per booking, reject overlaps, and support rescheduling.

4

Filling & Delivery

Record source, filled quantity, and delivery quantity within capacity limits, and flag delays.

5

Tariff & Billing

Maintain tariff slabs, generate bills using the applicable tariff, support partial payments, and mark overdue bills.

6

Complaints

Allow customers to raise complaints linked to bookings and track them from Open to Resolved.

7

Reporting

Provide reports on pending bookings, tanker utilization, delivery delays, source usage, customer history, complaints, and revenue.

4. Users & Stakeholders

Customer

- Register, add addresses, book, track, pay, complain

Dispatcher/Scheduler

- View bookings, assign tanker, driver, slot, avoid overlaps

Driver

- View schedule, confirm filling and delivery, report delays

Filling Station Operator

- Record filled quantity for the schedule

Billing Staff

- Apply tariff, generate bills, record payments, track overdue bills

Complaint Handler

- Log complaints, update status, resolve, close

Administrator

- Manage master data and reports

3. Objectives

01

Analyse requirements and identify entities, attributes, relationships and user roles

02

Design a conceptual (ER) model covering all 13 principal entities

03

Convert ER model into relational schema normalized to 3NF

04

Implement schema in DBMS with constraints reflecting business rules

05

Enforce critical rules: no overlapping schedules, capacity constraints, valid billing

06

Populate database with sample data and write SQL for operational reports

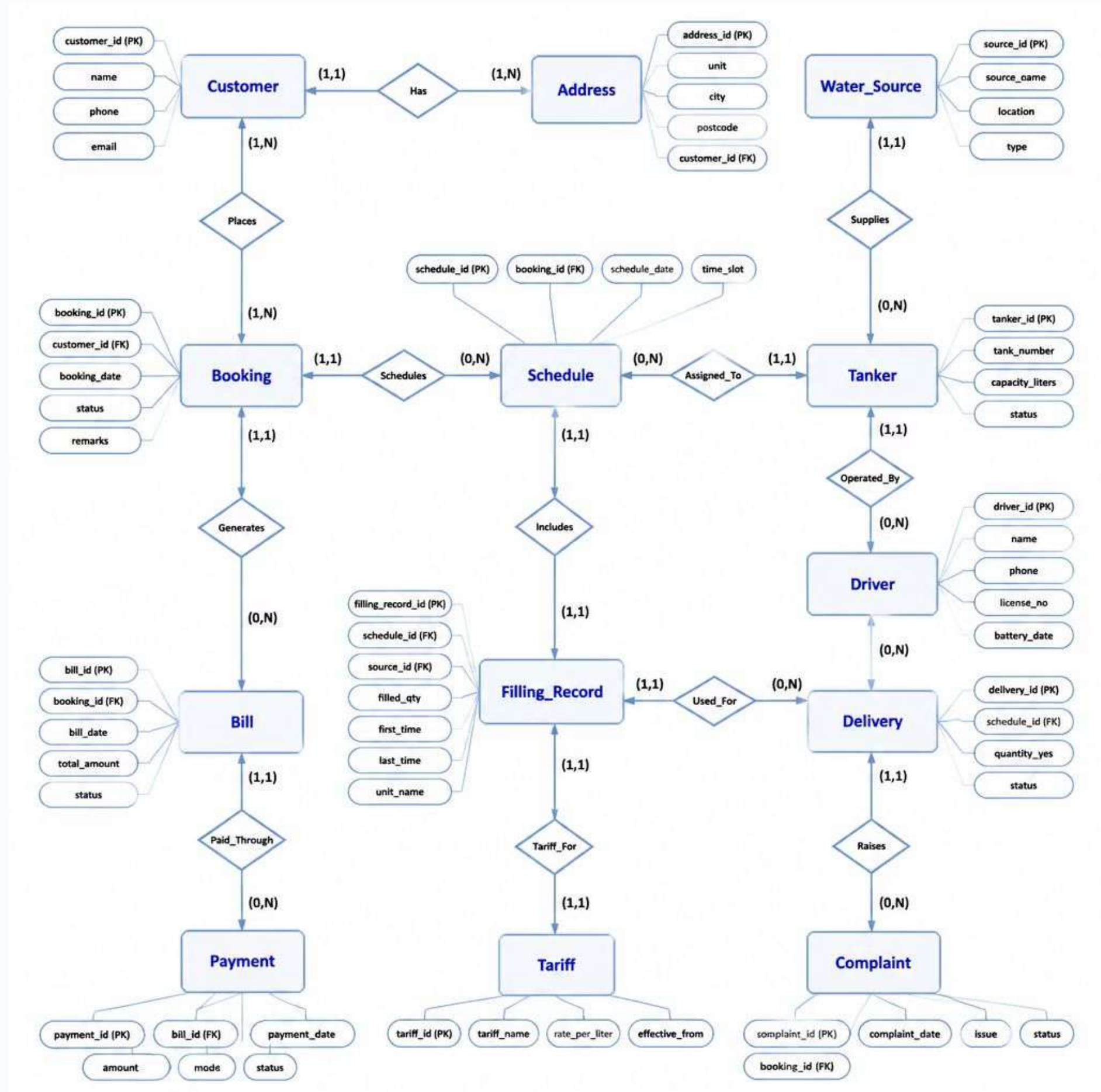
07

Build front-end (Java/Python) for booking, scheduling, billing and reporting

08

Demonstrate transaction awareness, indexing, and systematic testing

6. Conceptual Design – ER Diagram (13 Entities)



7. Entity Relationships & Cardinality

- **Customer** → many Addresses, Bookings, Complaints
- **Booking** → one Schedule, Filling_Record, Delivery, Bill
- **Schedule** → one Tanker, one Driver
- **Bill** → many Payments

8. Critical Business Rules

1

No Double-Booking

Prevent overlapping schedules by comparing booking date/time ranges against existing records for the same `tanker_id` and `driver_id`. Any conflict in the requested start/end window is rejected before insert or update.

2

Capacity Constraints

Enforce $filled_qty \leq tanker_capacity$ and $delivered_qty \leq filled_qty$ with database CHECK constraints. Invalid quantities are blocked at the table level, ensuring only feasible fuel movements are stored.

3

Billing Integrity

Calculate bill totals by applying the tariff rate to the delivered quantity, then adding any applicable tax or surcharge. Track each payment against the bill so cumulative payments cannot exceed the outstanding total.

9. Initial Relational Schema

The schema is designed to support the full tanker transport workflow with strong data integrity, minimal redundancy, and clear relationship enforcement across all entities.

→ Primary & Foreign Keys

All 13 tables use surrogate integer primary keys. Foreign keys link related records across customers, drivers, tankers, orders, schedules, deliveries, billing, and payments to enforce referential integrity.

→ CHECK & UNIQUE Constraints

CHECK constraints enforce business rules such as status values, date/time ordering, and quantity limits. UNIQUE constraints protect critical fields including phone numbers, email addresses, license numbers, and vehicle registration numbers.

→ Normalization

The schema is normalized to Third Normal Form (3NF) with no redundancy, ensuring each fact is stored once and all non-key attributes depend only on the key.